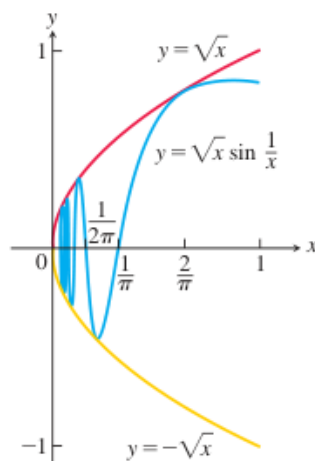


SIM1002 Calculus I
Tutorial 3
Limit of a Function and Limit Laws

Exercises 2.4

1. Let $g(x) = \sqrt{x} \sin\left(\frac{1}{x}\right)$



- (a) Does $\lim_{x \rightarrow 0^+} g(x)$ exist? If so, what is it? If not, why not?
- (b) Does $\lim_{x \rightarrow 0^-} g(x)$ exist? If so, what is it? If not, why not?
- (c) Does $\lim_{x \rightarrow 0} g(x)$ exist? If so, what is it? If not, why not?

Find the limits in Exercises Q2-Q5

2. $\lim_{x \rightarrow 1^+} \sqrt{\frac{x-1}{x-2}}$

3. (a) $\lim_{x \rightarrow -2^+} (x+3) \frac{|x+2|}{x+2}$

(b) $\lim_{x \rightarrow -2^-} (x+3) \frac{|x+2|}{x+2}$

4. (a) $\lim_{x \rightarrow 1^+} \frac{\sqrt{2x} |x-1|}{x-1}$

(b) $\lim_{x \rightarrow 1^-} \frac{\sqrt{2x} |x-1|}{x-1}$

5. (a) $\lim_{x \rightarrow 0^+} \frac{1 - \cos x}{|\cos x - 1|}$

(b) $\lim_{x \rightarrow 0^-} \frac{\cos x - 1}{|\cos x - 1|}$

Using $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ to solve the following.

6. $\lim_{y \rightarrow 0} \frac{\sin 3y}{4y}$

$$7. \quad \lim_{x \rightarrow 0} 6x^2(\cot x)(\operatorname{cosec} 2x)$$

$$8. \quad \lim_{x \rightarrow 0} \frac{x - x \cos x}{\sin^2 3x}$$

$$9. \quad \lim_{y \rightarrow 0} \frac{\sin 3y \cot 5y}{y \cot 4y}$$

Exercises 2.5

$$1. \quad \lim_{r \rightarrow \infty} \frac{r + \sin r}{2r + 7 - 5 \sin r}$$

$$2. \quad (a) \quad \lim_{x \rightarrow \infty} \frac{x^3 + 7x^2 - 2}{x^2 - x + 1}$$

$$(b) \quad \lim_{x \rightarrow -\infty} \frac{x^3 + 7x^2 - 2}{x^2 - x + 1}$$

$$3. \quad \lim_{x \rightarrow \infty} \sqrt{\frac{x^2 - 5x}{x^3 + x - 2}}$$

$$4. \quad \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 1}}{x + 1}$$

$$5. \quad \lim_{x \rightarrow -\infty} \frac{4 - 3x^3}{\sqrt{x^6 + 9}}$$

Infinite Limits

$$6. \quad \lim_{x \rightarrow 0} \frac{-1}{x^2(x+1)}$$

$$7. \quad \lim_{x \rightarrow 0} \frac{1}{x^{\frac{2}{3}}}$$

$$8. \quad \lim_{\theta \rightarrow 0} (2 - \cot \theta)$$

$$9. \quad \text{Find } \lim \left(\frac{1}{x^{\frac{1}{3}}} - \frac{1}{(x-1)^{\frac{4}{3}}} \right) \text{ as}$$

$$(a) \quad x \rightarrow 0^+$$

$$(b) \quad x \rightarrow 0^-$$

$$(c) \quad x \rightarrow 1^+$$

$$(d) \quad x \rightarrow 1^-$$

Domains, Ranges, and Asymptotes

Determine the domain and range of each function. Use various limits to find the asymptotes and the ranges.

1. $y = 4 + \frac{3x^2}{x^2+1}$

2. $y = \frac{2x}{x^2-1}$

3. $y = \frac{\sqrt{x^2+4}}{x}$

4. $y = \frac{x^3}{x^3-8}$

Find the limits in the following.

1. $\lim_{x \rightarrow \infty} (\sqrt{x+9} - \sqrt{x+4})$

2. $\lim_{x \rightarrow \infty} (\sqrt{x^2+3} - x)$

3. $\lim_{x \rightarrow \infty} (\sqrt{x^2+x} - \sqrt{x^2-x})$

Oblique Asymptotes

Graph the rational functions in the following. Include the graphs and equations of the asymptotes.

1. $y = \frac{x^2+1}{x-1}$

2. $y = \frac{x^2-1}{2x+4}$

3. $y = \frac{x^3+1}{x^2}$

Exercises 2.6

At what points are the functions in the following continuous?

1. $y = \frac{1}{|x|+1} - \frac{x^2}{2}$

2. $y = \frac{x+2}{\cos x}$

3. $y = \tan\left(\frac{\pi}{2}x\right)$

4. $y = \frac{\sqrt{x^4+1}}{1+\sin^2 x}$

5. $y = (2-x)^{\frac{1}{5}}$

Limits involving Trigonometric Functions

1. $\lim_{t \rightarrow 0} \sin\left(\frac{\pi}{2} \cos(\tan t)\right)$
2. $\lim_{t \rightarrow 0} \tan\left(\frac{\pi}{4} \cos\left(\sin x^{\frac{1}{3}}\right)\right)$
3. $\lim_{t \rightarrow 0} \sqrt{\operatorname{cosec}^2 x + 5\sqrt{3} \tan x}$

Continuous Extensions

1. For what value of b is

$$g(x) = \begin{cases} x, & x < -2 \\ bx^2, & x \geq -2 \end{cases}$$

continuous at every x ?

2. For what value of b is

$$g(x) = \begin{cases} \frac{x-b}{b+1}, & x \leq 0 \\ x^2 + b, & x > 0 \end{cases}$$

continuous at every x ?

3. For what value of a and b is

$$g(x) = \begin{cases} ax + 2b, & x \leq 0 \\ x^2 + 3a - b, & 0 < x \leq 2 \\ 3x - 5, & x > 2 \end{cases}$$

continuous at every x ?